In the Column Rebar Generator application, the ⚡ **Run** button and the 🔄 **Auto-Group Columns** button serve distinct roles in the workflow:

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1. ⚡ **Run** Button

- **What it does:** Performs a complete structural concrete design audit by extracting data directly from the active ETABS model and matching it against the AutoCAD schedule database.
- **Key Steps Executed:**
    1. Establishes a COM connection to the active ETABS session via etabs_connector.connect_to_etabs.
    2. Scans/extracts all column geometry, orientations, and concrete design reinforcement percentages via run_audit.
    3. Spatial Grouping: Spatially groups columns into vertical stack lines using plan coordinate keys: (grid_label, round(x, 3), round(y, 3)) via match_all_columns.
    4. Identifies the envelope maximum required rebar percentage and the largest physical cross-section size along each stack to query the CAD schedule database (Suggested_CAD_Rebar_Percentages.xlsx).
    5. Performs an initial pass of auto_group_columns to group them under schedule designations (e.g. C1, C2).
    6. Refreshes the HTML plan viewport and rebuilds the KPI dashboards.

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2. 🔄 **Auto-Group Columns** Button

- **What it does:** Instantly re-calculates the scheduling groups (notations C1, C2, etc.) on the already-extracted column data using the current value of the **Auto-Merge Tolerance** slider.
- **Key Steps Executed:**
    1. Reads the current matched column data in memory (does **not** query ETABS or reload files).
    2. Re-runs the grouping algorithm in run_auto_grouping calling auto_group_columns.
    3. Sorts columns of the same physical dimensions by their provided CAD steel percentage.
    4. Merges columns into the same schedule notation (e.g., C1) if their steel percentages differ by less than the chosen **Auto-Merge Tolerance** threshold (e.g., 0.30%).
    5. Enforces worst-case detailing: assigns the highest reinforcement requirement within each merged sub-group to all columns in that group for safety, updating their design efficiency metrics.
    6. Refreshes local UI filters, tables, and color-coded group notations.

Summary of Differences

Feature	⚡ **Run** Button	🔄 **Auto-Group Columns** Button
Data Source	Queries the live **ETABS model** and extracts structural results.	Uses **in-memory data** from the last active extraction.
Execution Time	Heavy (takes seconds; runs on a background thread).	Instantaneous (takes milliseconds; runs locally on the main thread).
Main Purpose	Imports new structural analysis data and performs initial matching.	Fine-tunes schedule groupings/labels to optimize drawing efficiency.
ETABS Needed?	**Yes** (must be connected to a running ETABS model).	**No** (runs entirely offline using local data).